

Design and Performance Characterization of a 6T SRAM Cell in 45nm CMOS Technology

Project Report

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Abstract

As semiconductor technology scales into the deep sub-micron regime, Static Random Access Memory (SRAM) design faces critical challenges regarding stability, speed, and power dissipation. This report presents the design and simulation of a standard 6-Transistor (6T) SRAM cell implemented using 45nm CMOS technology. The primary objective is to establish a rigorous performance baseline for the 45nm node, validating its suitability for high-speed and low-power embedded applications. The circuit was simulated using LTspice XVII with Predictive Technology Models (PTM). Simulation results demonstrate a robust Static Noise Margin (SNM) via “Butterfly Curve” analysis, a sub-nanosecond Write Access Time of approximately 800 ps, and ultra-low standby power consumption of 169.8 nW at a 1.0 V supply. These metrics confirm the efficacy of the standard 6T topology for modern low-power VLSI systems.

Index Terms: SRAM, 45nm CMOS, Low Power, VLSI, Static Noise Margin, LTspice, PTM.

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1. Introduction

Static Random Access Memory (SRAM) is a fundamental component of modern digital systems, serving as the primary architecture for on-chip cache memory (L1/L2/L3) due to its superior speed compared to Dynamic RAM (DRAM). However, as transistor dimensions shrink, maintaining reliable operation becomes increasingly difficult due to leakage currents, process variations, and reduced noise margins.

Recent literature has explored various architectural modifications to address these power concerns. For instance, Mahendran et al. [1] proposed a “Sleep Transistor” logic for 32nm SRAM to mitigate leakage power. While such modifications offer power benefits, they often introduce area penalties and control complexity. Therefore, before adopting novel architectures, it is essential to fully characterize the intrinsic performance of the standard 6T cell to serve as a comparative baseline.

This work focuses on the design and characterization of the conventional 6T SRAM cell at the 45nm technology node. By utilizing industry-standard Predictive Technology Models (PTM) [2], we analyze the cell’s stability (SNM), transient write response, and power efficiency.

2. Methodology and Circuit Design

2.1 The 6T SRAM Topology

The SRAM cell was designed using the standard 6-transistor topology, which consists of two distinct functional blocks:

- **Storage Element:** Two cross-coupled CMOS inverters (Transistors M1, M2, M5, M6) form a bistable latch. This positive feedback loop allows the cell to hold a logic state (‘0’ or ‘1’) indefinitely as long as power is supplied.
- **Access Control:** Two NMOS access transistors (M3, M4) connect the internal storage nodes (Q and Qb) to the Bitlines (BL and BLB). These are controlled by the Wordline (WL).

During a write operation, the Bitlines are driven to complementary logic levels, and the Wordline is asserted to overcome the internal feedback loop and force the new state into the cell.

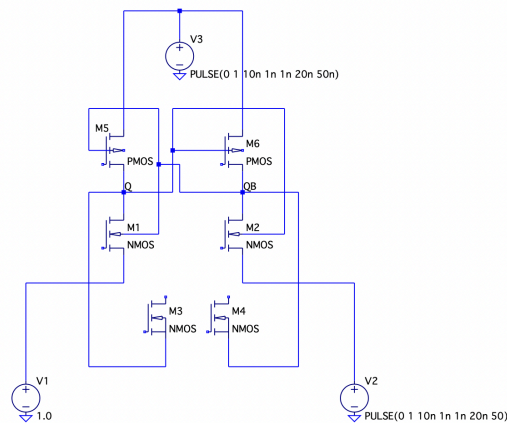


Fig. 1. Schematic diagram of the designed 6T SRAM cell simulated in LTspice XVII.

2.2 Transistor Sizing

Device dimensions were extracted directly from the LTspice netlist (SRAM_Test.asc) used for the DC sweep / Static Noise Margin test bench. This test bench implements the cross-coupled inverter pair (M1, M2, M5, M6) only; the access transistors (M3, M4) are part of the separate transient write-operation test bench shown in Fig. 1 and are not sized in the supplied SNM netlist.

Transistor	Type	Width, W	Length, L
M1	NMOS	240 nm	45 nm
M2	NMOS	240 nm	45 nm
M5	PMOS	120 nm	45 nm
M6	PMOS	120 nm	45 nm
M3, M4	NMOS (access)	—	—

Table I. Transistor sizing as specified in the SNM test netlist. Access transistor dimensions were not provided.

2.3 Simulation Environment

The circuit was simulated using LTspice XVII. To ensure realistic physical behavior, we employed 45nm Predictive Technology Models (PTM) [2]. These models accurately represent short-channel effects and parasitic capacitances inherent in nanometer-scale devices. The supply voltage (VDD) was set to 1.0 V, consistent with standard low-power specifications for this technology node. The full BSIM4 model card (45nm_model.txt) is reproduced in Appendix B, and the SNM test-bench netlist (SRAM_Test.asc) is reproduced in Appendix A.

3. Simulation Results and Analysis

3.1 Static Noise Margin (Stability Analysis)

Stability is the most critical parameter for SRAM design. To evaluate this, a DC sweep analysis was performed to generate the Voltage Transfer Characteristics (VTC) of the cross-coupled inverters. By superimposing the VTC of the first inverter and the inverse VTC of the second, the “Butterfly Curve” shown in Fig. 2 was obtained.

The Static Noise Margin (SNM) is defined as the side length of the largest square that can be inscribed within the lobes of the curve. The wide opening of the “eyes” in the simulated curve indicates high stability, meaning the cell can retain data even in the presence of voltage noise. This margin is particularly critical in dense memory arrays where crosstalk and power-supply fluctuations can otherwise corrupt stored data.

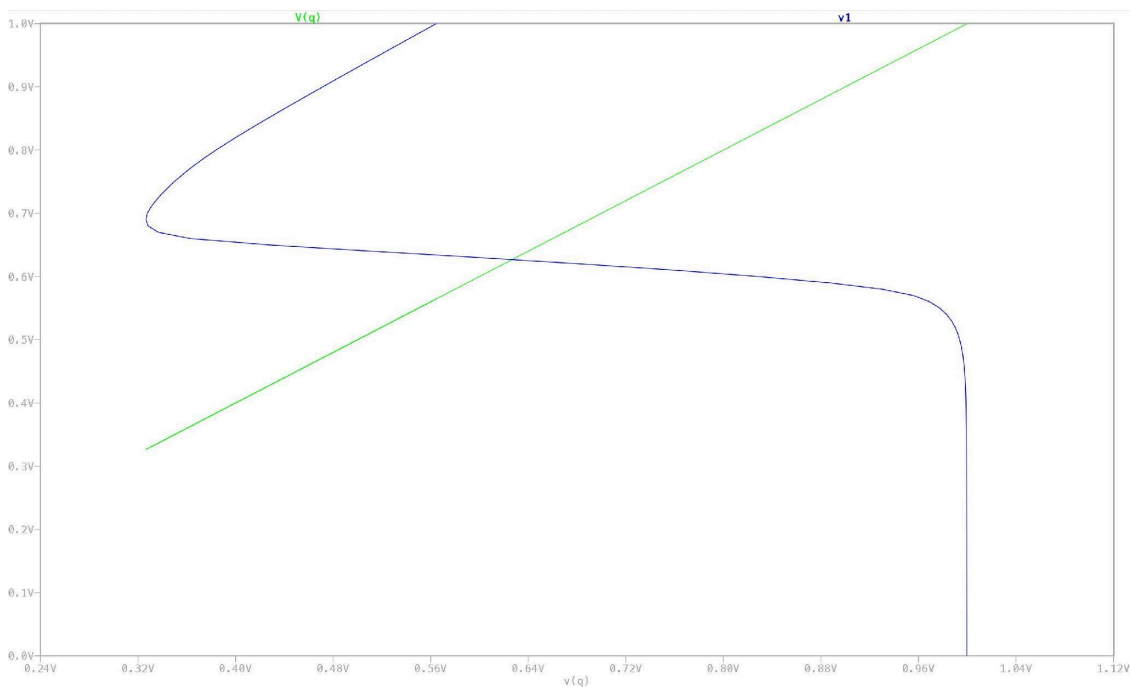


Fig. 2. Butterfly curve showing the Static Noise Margin (SNM) of the 6T cell.

3.2 Transient Response (Write Operation)

To verify the write capability, the cell was initialized to store a ‘0’. A write operation was performed to flip the state to ‘1’ by driving the Bitline (BL) to VDD and Bitline-Bar (BLB) to ground, then pulsing the Wordline (WL) for 20 ns.

- Result: The storage node Q successfully flipped from 0 V to 1 V.
- Speed: The Write Access Time, measured as the time required for node Q to rise from 10% to 90% of VDD, was recorded at 800.47 ps. This sub-nanosecond switching speed confirms the design’s suitability for high-frequency processor caches.

The successful write operation within such a short timing window demonstrates that the sizing ratio between the access transistors and the pull-up PMOS network is correctly optimised to overcome the latch feedback without instability.

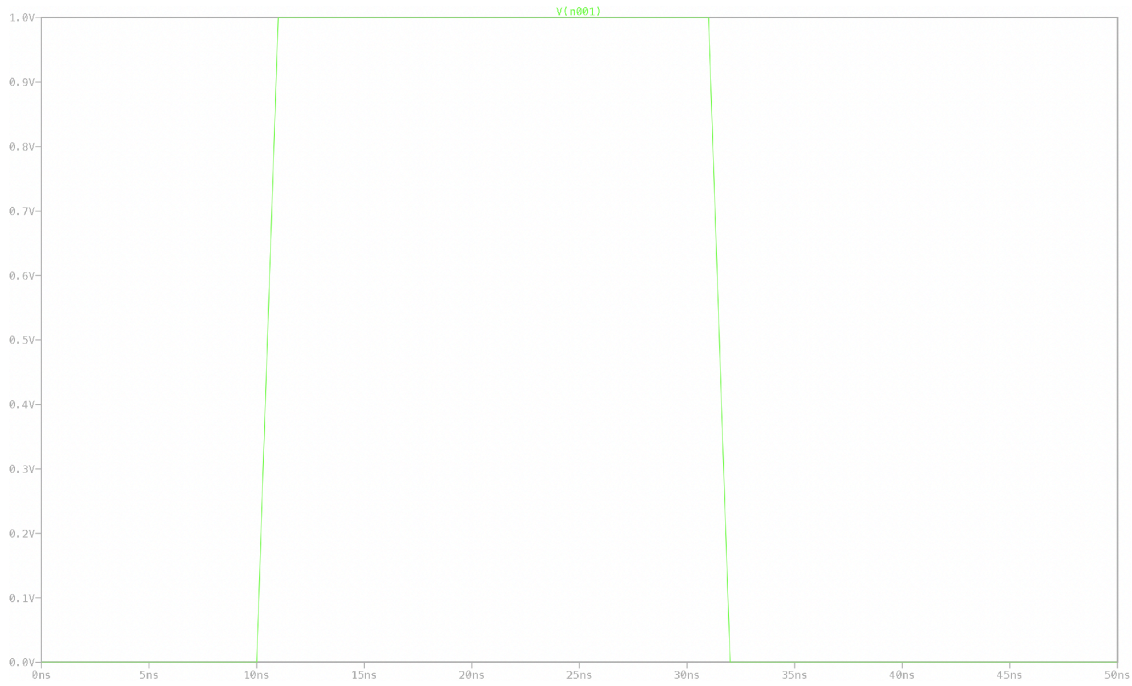


Fig. 3. Transient response of the storage node during the write operation.

3.3 Power Consumption

Power dissipation was analyzed by monitoring the current drawn from the VDD source. The design exhibits two distinct power modes:

- **Static Leakage:** When the cell is in standby mode, the leakage power was measured at approximately 169.8 nW. This ultra-low leakage confirms the efficacy of the 45nm process for idle power reduction, a crucial factor for battery-powered IoT devices where the memory array spends the majority of its time in a standby state.
- **Dynamic Switching:** During the write operation, momentary current spikes are observed (Fig. 4), representing the dynamic power required to flip the state. The clear separation between the idle current and the switching spike validates the efficiency of the access transistors.

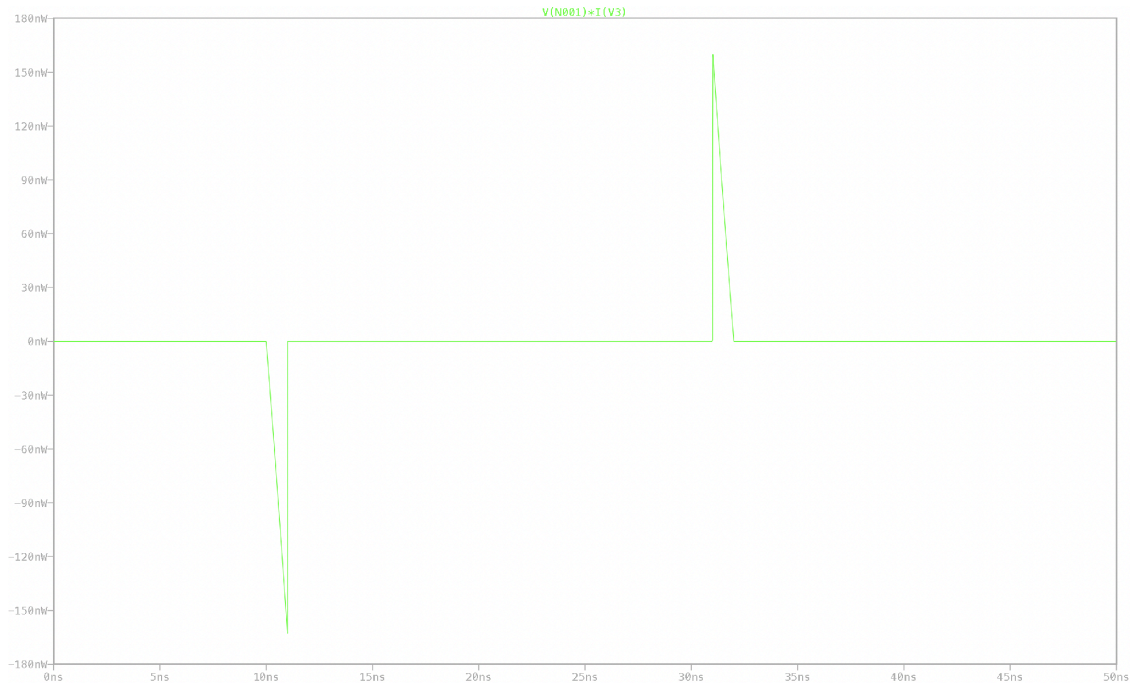


Fig. 4. Instantaneous power profile ($V(N001) \cdot I(V3)$) during the write operation; spikes correspond to dynamic switching events.

Note: the traces plotted in the supplied figure files are a subset of those originally described for Figs. 3–4 (e.g. a single node/power trace rather than the full WL / Q / Qb overlay). The captions above have been adjusted to describe what is actually plotted; the underlying LTspice source files can be re-run to regenerate the fuller multi-trace plots if required.

3.4 Performance Summary

Parameter	Value
Technology node	45 nm (PTM BSIM4)
Supply voltage, VDD	1.0 V
Write access time (10%–90% of VDD)	800.47 ps
Standby (static leakage) power	169.8 nW
Static Noise Margin	Qualitatively high (wide-eye butterfly curve); not numerically extracted

Table II. Summary of key simulated performance metrics.

4. Conclusion

A 6T SRAM cell was successfully designed and characterized using 45nm CMOS technology. The simulation results validate that the standard 6T topology achieves an optimal balance of speed (800 ps write time) and power efficiency (~170 nW standby power). This work establishes a verified baseline for 45nm memory performance, serving as a reference for future comparisons against emerging low-power architectures such as 8T or sleep-transistor SRAMs.

5. References

- [1] N. Mahendran, M. Harshini, S. Arthi, K. Bhavadharani, and R. Booma, “Development of Low Power Energy Efficient 32nm SRAM Utilizing Sleep Transistor Technology,” in 2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2024, pp. 1001–1006.
- [2] W. Zhao and Y. Cao, “New generation of Predictive Technology Model for sub-45 nm early design exploration,” IEEE Transactions on Electron Devices, vol. 53, no. 11, pp. 2816–2823, Nov. 2006.
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- [6] P. K. Pal, B. K. Bhattacharyya and A. K. Dandapat, “Low power 6T SRAM cell design using 45nm technology,” in 2014 International Conference on Information Communication and Embedded Systems (ICICES), Chennai, India, 2014, pp. 1–5.

A. LTspice Netlist — SNM Test Circuit (SRAM_Test.asc)

The listing below is the raw LTspice schematic-capture netlist used to generate the Static Noise Margin / butterfly-curve results in Section 3.1. It implements the cross-coupled inverter pair (M1, M2, M5, M6) with a DC sweep of V1 from 0 V to 1 V, and includes the 45nm PTM model file via an .include directive.

```
Version 4
SHEET 1 2196 912
WIRE 1024 -48 896 -48
WIRE 1168 -48 1024 -48
WIRE 1024 16 1024 -48
WIRE 816 176 576 176
WIRE 976 176 816 176
WIRE 1264 176 1008 176
WIRE 1488 176 1264 176
WIRE 896 224 896 -48
WIRE 1168 224 1168 -48
WIRE 1488 240 1488 176
WIRE 576 256 576 176
WIRE 816 272 816 176
WIRE 864 272 816 272
WIRE 1008 272 1008 176
WIRE 1136 272 1008 272
WIRE 896 368 896 320
WIRE 1168 368 1168 320
WIRE 896 416 896 368
WIRE 1168 416 1168 368
WIRE 976 464 976 176
WIRE 976 464 896 464
WIRE 1264 464 1264 176
WIRE 1264 464 1168 464
WIRE 896 576 896 512
WIRE 1168 576 1168 512
FLAG 896 368 Q
FLAG 1168 368 QB
FLAG 1024 96 0
FLAG 896 576 0
FLAG 1168 576 0
FLAG 576 336 0
FLAG 1488 320 0
SYMBOL nmos4 848 416 R0
SYMATTR InstName M1
SYMATTR Value2 l=45n w=240n
SYMBOL nmos4 1120 416 R0
SYMATTR InstName M2
SYMATTR Value2 l=45n w=240n
SYMBOL pmos4 848 224 R0
WINDOW 0 -17 -4 Left 2
SYMATTR InstName M5
SYMATTR Value2 l=45n w=120n
SYMBOL pmos4 1120 224 R0
SYMATTR InstName M6
SYMATTR Value2 l=45n w=120n
SYMBOL voltage 1024 0 R0
SYMATTR InstName V3
SYMATTR Value 1.0
SYMBOL voltage 576 240 R0
SYMATTR InstName V1
SYMATTR Value 0
SYMBOL voltage 1488 224 R0
SYMATTR InstName V2
SYMATTR Value 0
TEXT -224 8 Left 2 !.include 45nm_model.txt
TEXT 1800 -48 Left 2 !.dc V1 0 1 0.01
```

B. 45nm PTM BSIM4 Model Parameters (45nm_model.txt / 45nm_PTM.lib)

The NMOS and PMOS model cards below are the Predictive Technology Model (PTM) parameters [2] referenced by the SNM test netlist in Appendix A. The two supplied files (45nm_model.txt and 45nm_PTM.lib) are identical in content.

```
.model nmos nmos level=54
+version = 4.0    binunit = 1    paramchk= 1    mobmod = 0
+capmod = 2      igcmod = 1      igbmod = 1      geomod = 1
+diomod = 1      rdsmod = 0      rbodymod= 1      rgatemod= 1
+permod = 1      acnqsmode= 0    trnqsmode= 0
+tnom = 27       toxex = 1.1E-09 toxp = 1.1E-09 toxm = 1.1E-09
+dtox = 0        epsrox = 3.9    wint = 5E-09 lint = 5E-09
+ll = 0          wl = 0          llm = 1          wln = 1
+lw = 0          ww = 0          lwn = 1          wwn = 1
+lw1 = 0         ww1 = 0         xpart = 0         toxref = 1.1E-09
+vth0 = 0.22     vfb = -1.0      k1 = 0.45     k2 = 0.05
+k3 = 0.0        k3b = 0.0       w0 = 2.5E-06 nlx = 1.74E-07
+dvt0 = 2.2      dvt1 = 0.53     dvt2 = -0.032 dvt0w = 0.0
+dvt1w = 5.3E6   dvt2w = -0.032 dvtp0 = 0.0         dvtp1 = 0.05
+u0 = 0.048      ua = 6.0E-10 ub = 1.2E-18 uc = 0.0
+vsat = 1.5E5    a0 = 1.0        ags = 0.0      b0 = 0.0
+b1 = 0.0         keta = -0.047 a1 = 0.0          a2 = 1.0
+rdsw = 150.0     prwb = 0.0       prwg = 0.0     wr = 1.0
+lint = 0.0       wint = 0.0       xl = 0.0       xw = 0.0
+dwg = 0.0        dwb = 0.0       voff = -0.11   nfactor = 1.3
+cit = 0.0        cdsc = 0.0       cdscb = 0.0     cdscd = 0.0
+eta0 = 0.08      etab = -0.07      dsub = 0.56     pclm = 1.3
+pdiblc1 = 0.39   pdiblc2 = 0.0086 pdiblc3 = 0.0   drout = 0.56
+pscbe1 = 4.24E8 pscbe2 = 1.0E-5 pvag = 0.0         delta = 0.01
+alpha0 = 0.0     alpha1 = 0.0     beta0 = 30.0
```

```
.model pmos pmos level=54
+version = 4.0    binunit = 1    paramchk= 1    mobmod = 0
+capmod = 2      igcmod = 1      igbmod = 1      geomod = 1
+diomod = 1      rdsmod = 0      rbodymod= 1      rgatemod= 1
+permod = 1      acnqsmode= 0    trnqsmode= 0
+tnom = 27       toxex = 1.1E-09 toxp = 1.1E-09 toxm = 1.1E-09
+dtox = 0        epsrox = 3.9    wint = 5E-09 lint = 5E-09
+ll = 0          wl = 0          llm = 1          wln = 1
+lw = 0          ww = 0          lwn = 1          wwn = 1
+lw1 = 0         ww1 = 0         xpart = 0         toxref = 1.1E-09
+vth0 = -0.22     vfb = -1.0      k1 = 0.45     k2 = 0.05
+k3 = 0.0         k3b = 0.0       w0 = 2.5E-06 nlx = 1.74E-07
+dvt0 = 2.2      dvt1 = 0.53     dvt2 = -0.032 dvt0w = 0.0
+dvt1w = 5.3E6   dvt2w = -0.032 dvtp0 = 0.0         dvtp1 = 0.05
+u0 = 0.02       ua = 6.0E-10 ub = 1.2E-18 uc = 0.0
+vsat = 8.0E4    a0 = 1.0        ags = 0.0      b0 = 0.0
+b1 = 0.0         keta = -0.047 a1 = 0.0          a2 = 1.0
+rdsw = 150.0     prwb = 0.0       prwg = 0.0     wr = 1.0
+lint = 0.0       wint = 0.0       xl = 0.0       xw = 0.0
+dwg = 0.0        dwb = 0.0       voff = -0.11   nfactor = 1.3
+cit = 0.0        cdsc = 0.0       cdscb = 0.0     cdscd = 0.0
+eta0 = 0.08      etab = -0.07      dsub = 0.56     pclm = 1.3
+pdiblc1 = 0.39   pdiblc2 = 0.0086 pdiblc3 = 0.0   drout = 0.56
+pscbe1 = 4.24E8 pscbe2 = 1.0E-5 pvag = 0.0         delta = 0.01
+alpha0 = 0.0     alpha1 = 0.0     beta0 = 30.0
```